Adit Sanghani

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EDUCATION

Texas A&M University, *BS in Mechanical Engineering (GPA: 3.6/4.0)* *Aug '21 - May '25*

ACHIEVEMENTS

- Inventor on patent-pending robotic leak isolation system (in collaboration with Dell Technologies)
- Published author in Advanced Materials Technologies (Preferential Contamination in Electroadhesive Touchscreens)
- Dean's Honor Roll, Distinguished Student Award, Aggieland Bound Scholar at Texas A&M University.

WORK EXPERIENCE

Nucor Steel Texas, *Process Automation Engineer* *Jun '25 - Present*

- Led design and deployment of electromechanical system upgrades for legacy industrial equipment, integrating closed-loop control and process intelligence to improve performance, reliability, and system-level behavior.
- Designed and deployed a closed-loop pneumatic actuation system, integrating sensors, valves, and control logic to eliminate manual operation; delivered ~$200K in annual cost savings.
- Developed and implemented a computer vision metrology system, eliminating manual measurement, improving accuracy by ~30%, and reducing operator involvement to near zero.

Tesla, *Equipment Design Intern* *Aug '23 - Jan '24; May '24 – Aug '24*

- Drove automation and robotic system upgrades for new product introduction (NPI) and to improve factory-wide uptime and throughput.
- Engineered an adaptive mechanical support system for AGV fleets using sensor-driven feedback for real-time kinematic adjustments; eliminated the need for 1 maintenance headcount per shift (~$344K annual savings).
- Architected a $75K cost-effective automated conveyor system to enable seamless interface between forklifts and AGVs; established as the factory standard and replicated across 6+ production lines.

Dell Technologies, *Project Lead* *Aug '24 – May '25*

- Led end-to-end development of a patent-pending robotic leak isolation system for liquid-cooled server racks, mitigating $1M+ potential downtime and hardware damage per failure event.
- Developed sensor-driven automation and control logic to monitor rack conditions, actuate valves, and reroute coolant through redesigned flow architecture and within 15 seconds of detection.
- Conducted system-level validation across L11 server infrastructure, ensuring mechanical scalability and long-term reliability in high-density data center environments, demonstrated 95% leak detection accuracy.

INVENT Lab, Texas A&M University, *Engineering Technician* *May '22 – May '25*

- Created turnkey electromechanical research equipment to advance the development of dexterous humanoid hands as part of the NSF HAND ERC consortium.
- Integrated precision mechatronic platforms, combining actuators, sensors, and motion control hardware to automate testing of robotic contact and grasp behaviors.
- Prototyped and validated robotic end-effectors and subassemblies using SolidWorks, MATLAB, and LabVIEW, improving repeatability and reducing test cycle time.

SKILLS

Design: NX, SolidWorks, Creo, CATIA, AutoCAD, Onshape, GD&T, Tolerance Analysis, DFMA, Mechanism Design
Automation & Controls: MATLAB, LabVIEW, Python, Sensors & Actuators, Motion Control, Feedback Loops, FMEA
Manufacturing & Prototyping: CNC, Rapid Prototyping, Sheet Metal, Weldments, Machined Components, Precision Assembly
Quality & Data Analysis: ISO 9001/14001, Process Capability (Cp/Cpk), GR&R, SPC, DOE, Six Sigma, RCA, FEA